The Unreasonable Effectiveness of Quarto

John Paul Helveston

2026 Quant UX Conference

Handwriting practice sheets



Prompt:

*“Make a one-page handwriting practice sheet with sentences about **pokemon**”*

Pokémon Handwriting Practice

Trace the sentence, then practice writing it on your own.

Pokemon are amazing creatures to catch.

Pikachu is the most famous Pokemon.

There are over nine hundred Pokemon now.

Pokemon battle each other in gyms.

Some Pokemon evolve into stronger forms.

Charizard is a fire and flying type.

You need all eight gym badges to win.


8 Rounds Later...

Minecraft – Handwriting Practice

Minecraft is a popular building game.

Players mine for resources underground.

You can build almost anything you want.

Creepers are sneaky and dangerous enemies.

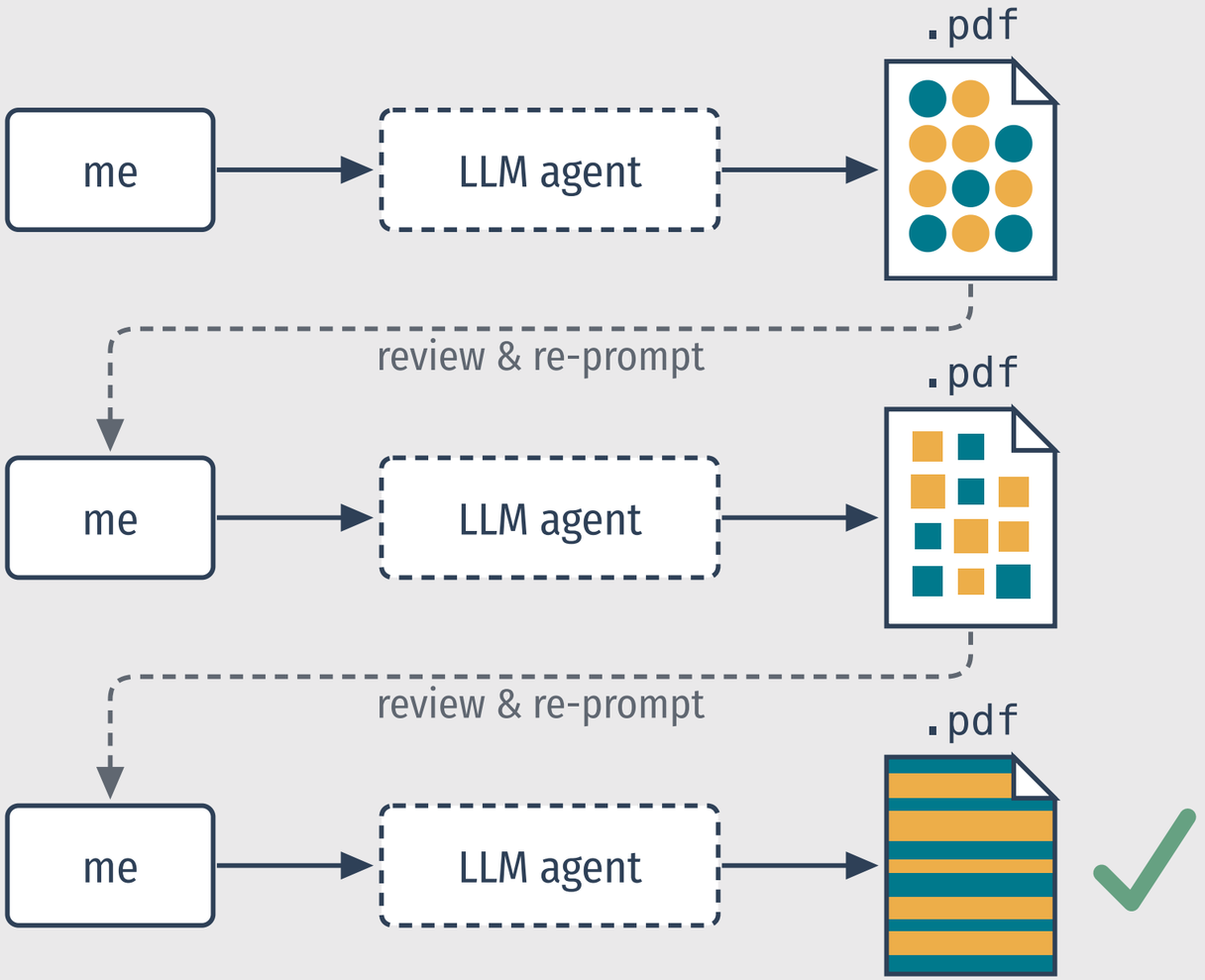
Diamonds are the best ore to find.

Players must survive through the first night.

The Nether is a hot and scary place.

I want **these** sentences with **this** layout

Content entangled with Format

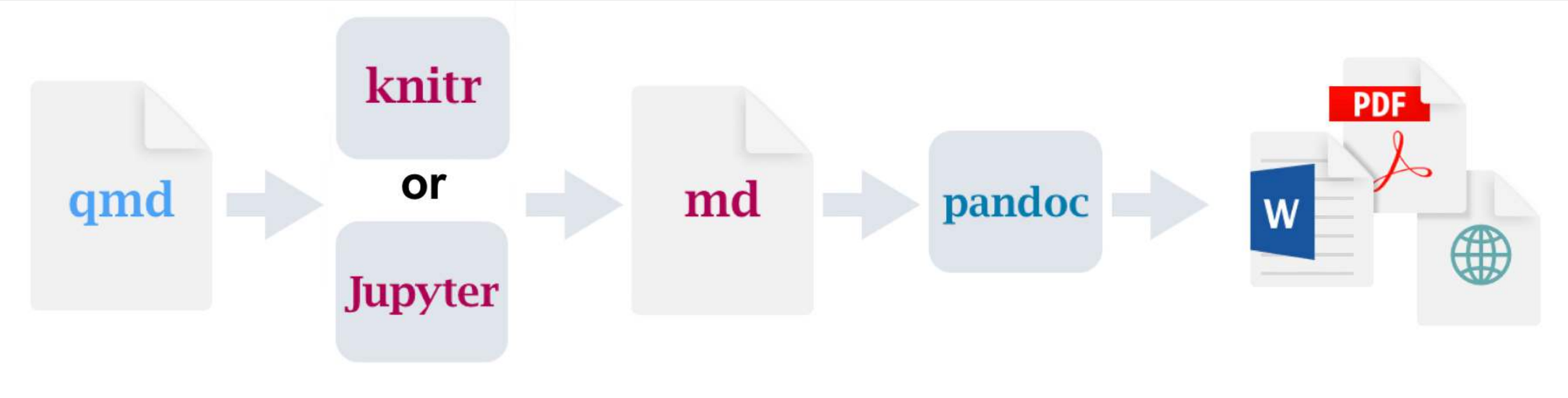


Quarto disentangles

Content and Format

So what *is* Quarto?

Quarto is a publishing system



Example `.qmd` file

```
---
format: html
title: "HTML Page with R Code"
---

# Hello, World!

This is a simple **HTML** page with *R* code.

```{r}
library(ggplot2)

df <- data.frame(x = rnorm(100))
ggplot(df, aes(x = x)) +
  geom_histogram()
```
```

Format

What it *becomes*

Content

What it *says*

Rendered: Markdown + R code chunks

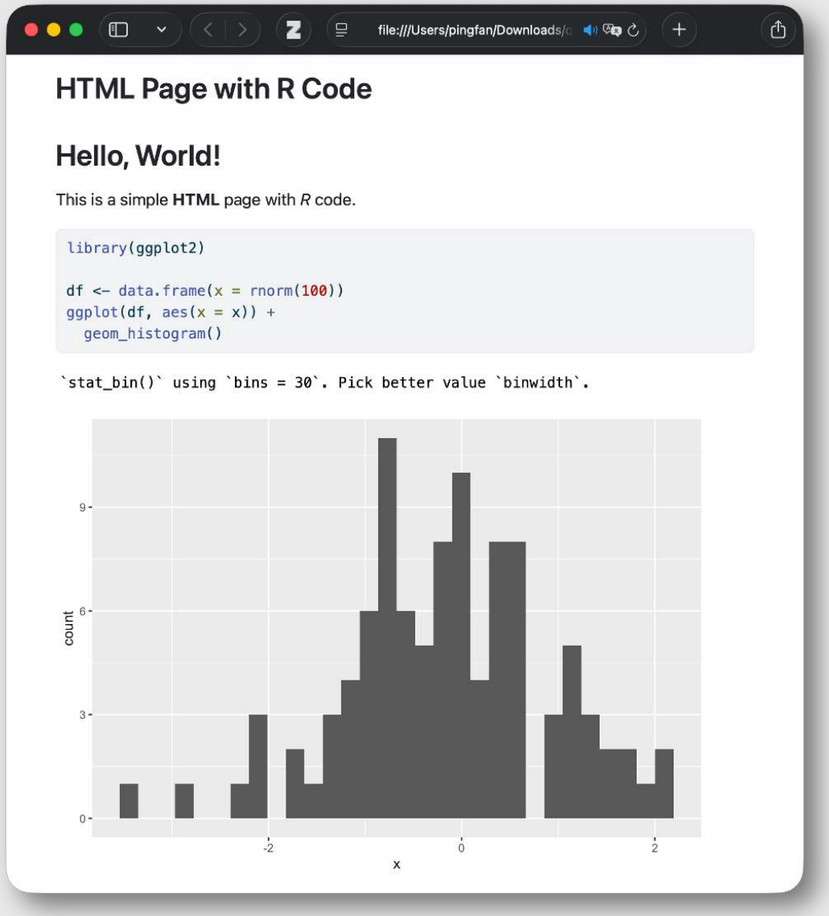
````
---
format: html
title: "HTML Page with R Code"
---

# Hello, World!

This is a simple **HTML** page with *R* code.

```{r}
library(ggplot2)

df <- data.frame(x = rnorm(100))
ggplot(df, aes(x = x)) +
  geom_histogram()
```
````



Rendered: Markdown + Python code chunks

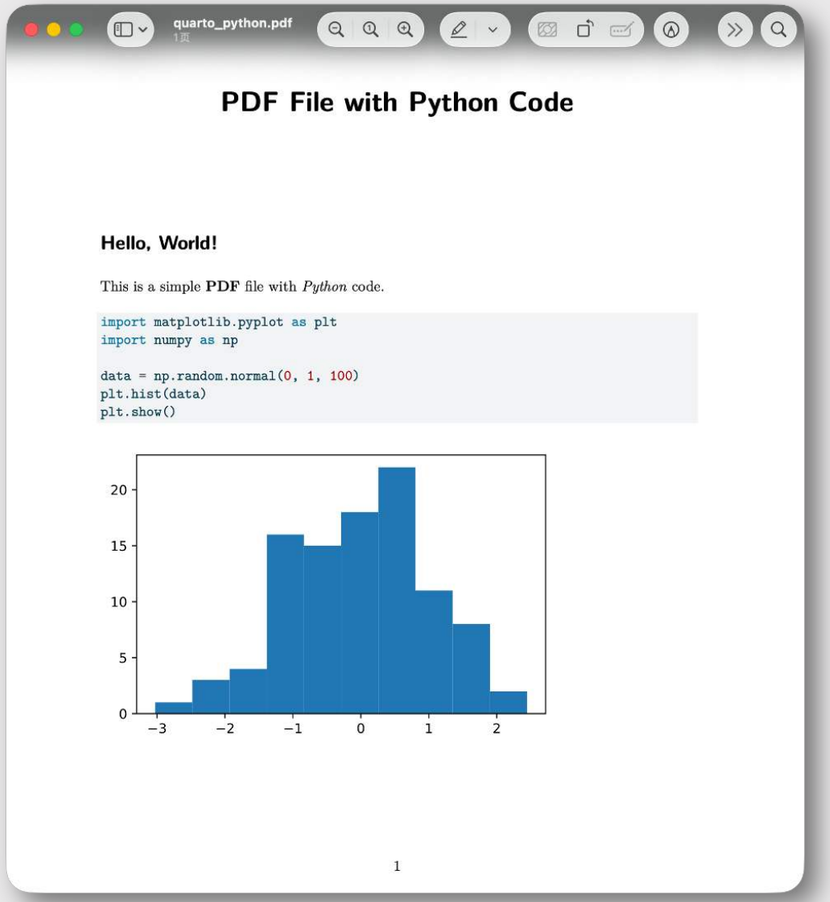
````
---
format: pdf
title: "PDF File with Python Code"
---

# Hello, World!

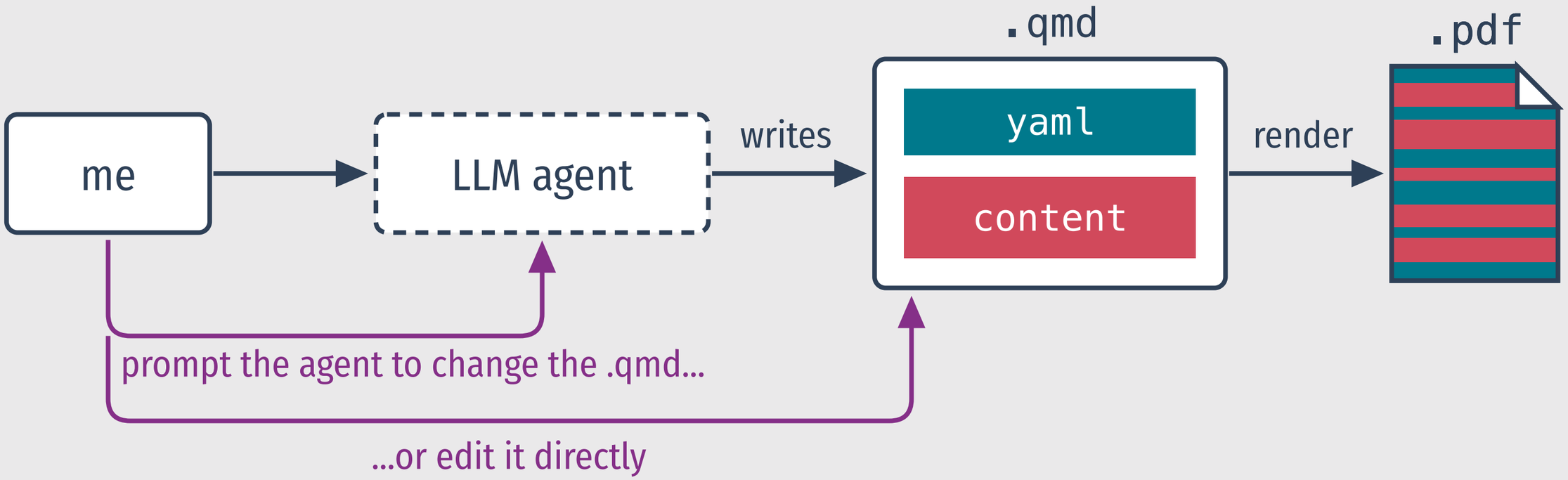
This is a simple **PDF** file with *Python* code.

```{python}
import matplotlib.pyplot as plt
import numpy as np

data = np.random.normal(0, 1, 100)
plt.hist(data)
plt.show()
```
````



Quarto disentangles Content and Format



Modularized content via parameters

handwriting-practice.qmd

(written by Claude Code)

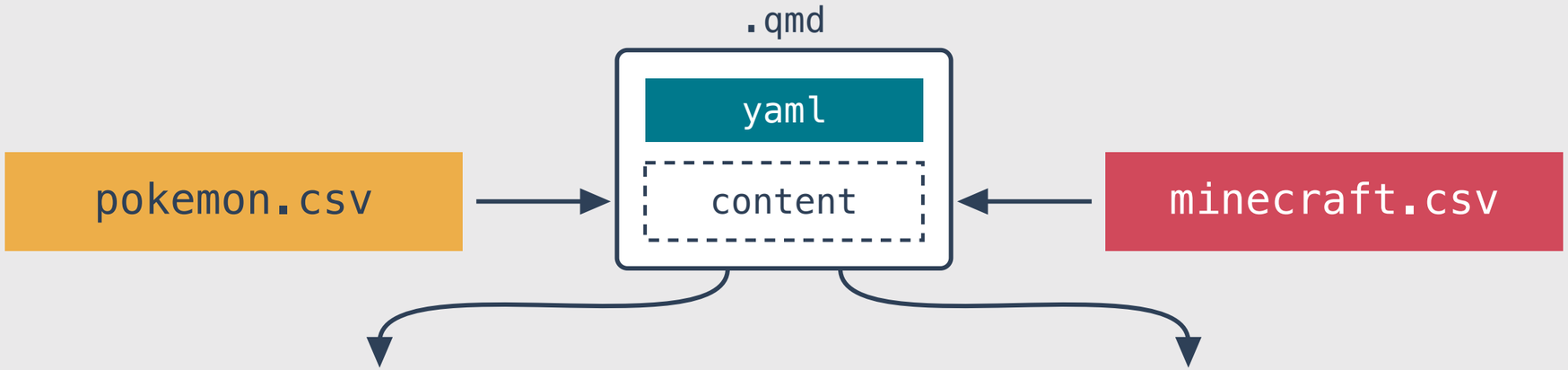
````
---
params:
  data_file: "data/minecraft.csv"
format: pdf
---

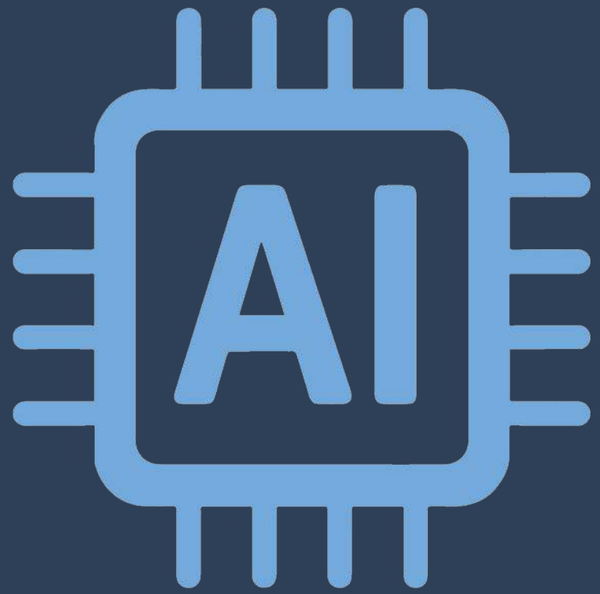
```{r}
sentences <- read.csv(params$data_file)$sentence
for (s in sentences) {
  cat(sprintf("\\practiceblock{%s}\n\n", s))
}
```
````

minecraft.csv

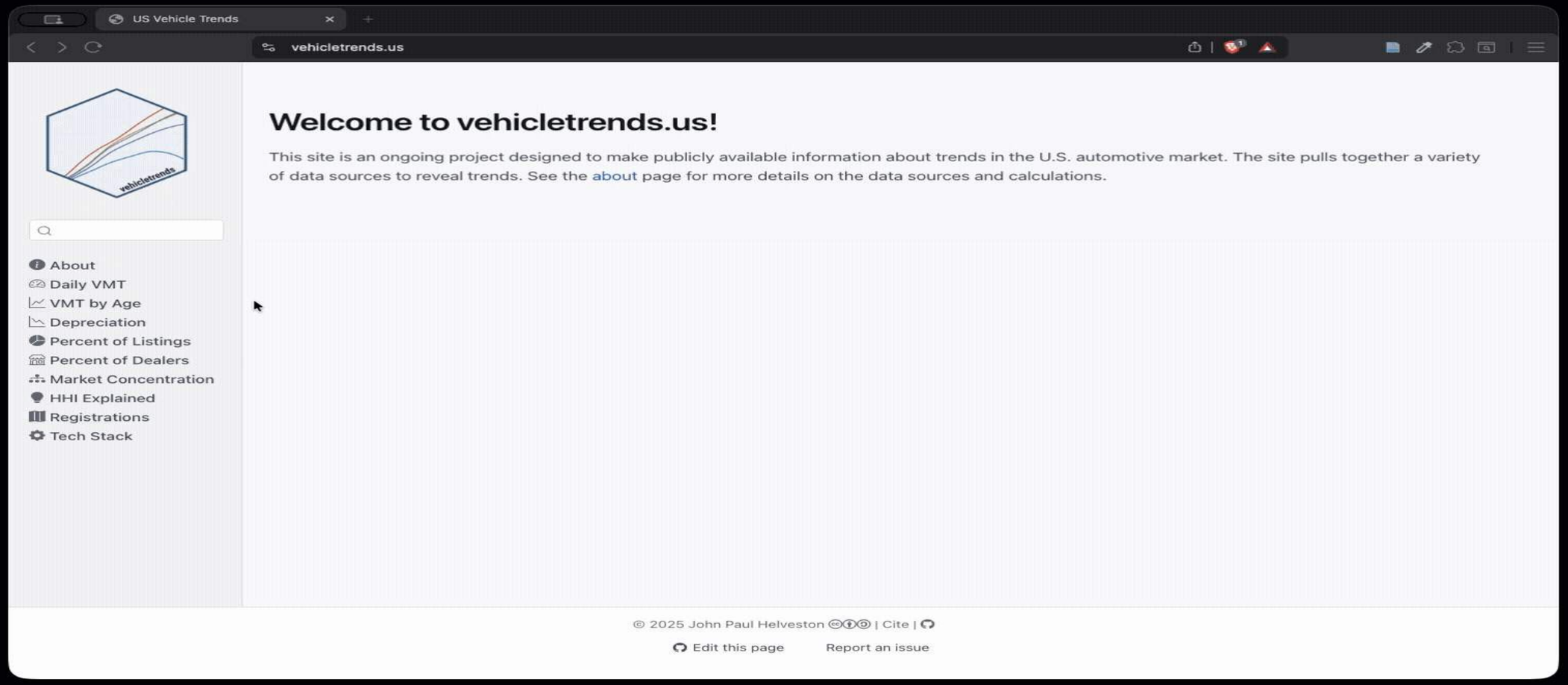
(written by Me)

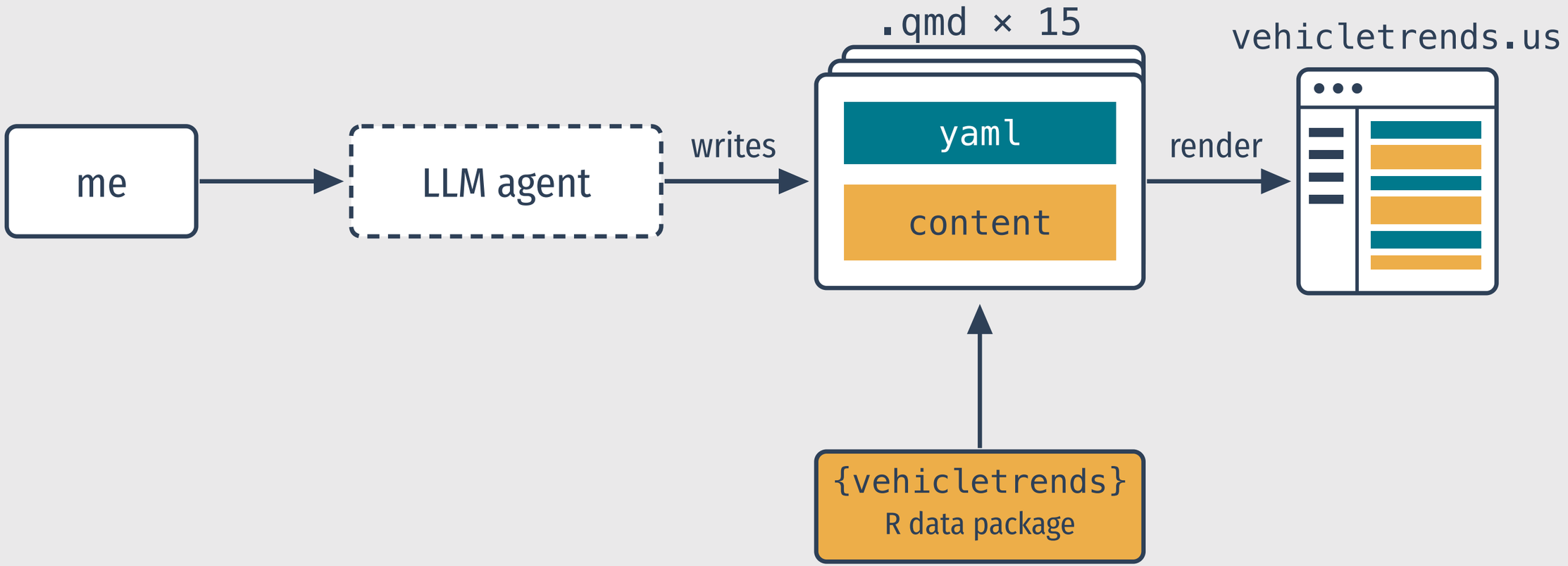
sentence
Minecraft is a popular building game.
Players mine for resources underground.
You can build almost anything you want.
Creepers are sneaky and dangerous enemies.
Diamonds are the best ore to find.
Players must survive through the first night.
The Nether is a hot and scary place.

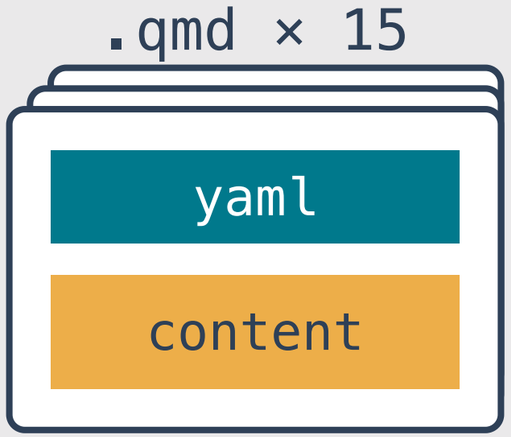

.qmd
yaml
content
pokemon.csv
minecraft.csv


AI


Epic Dashboards
AI


US Vehicle Trends
vehicletrends.us
vehicletrends
Welcome to vehicletrends.us!
This site is an ongoing project designed to make publicly available information about trends in the U.S. automotive market. The site pulls together a variety of data sources to reveal trends. See the about page for more details on the data sources and calculations.
About
Daily VMT
VMT by Age
Depreciation
Percent of Listings
Percent of Dealers
Market Concentration
HHI Explained
Registrations
Tech Stack
© 2025 John Paul Helveston | Cite |
Edit this page
Report an issue

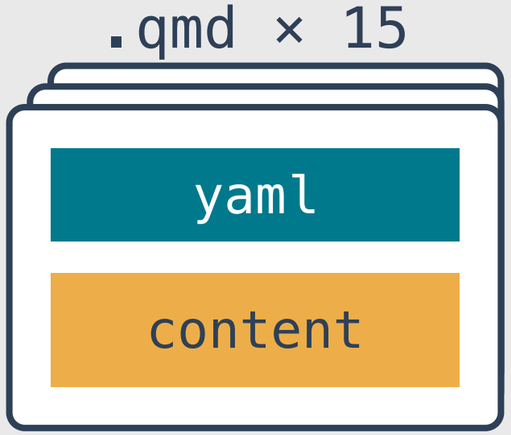

me
LLM agent
writes
.qmd × 15
yaml
content
render
vehicletrends.us
{vehicletrends}
R data package



_quarto.yml

```yaml
project:
  type: website

website:
  sidebar:
    contents:
      - about.qmd
      - vmt-daily.qmd
      - vmt-age.qmd
      - depreciation.qmd
      - ...
```



_quarto.yml

```yaml
project:
  type: website

website:
  sidebar:
    contents:
      - about.qmd
      - vmt-daily.qmd
      - vmt-age.qmd
      - depreciation.qmd
      - ...
```

Charts built with {vehicletrends}

A cohesive framework:

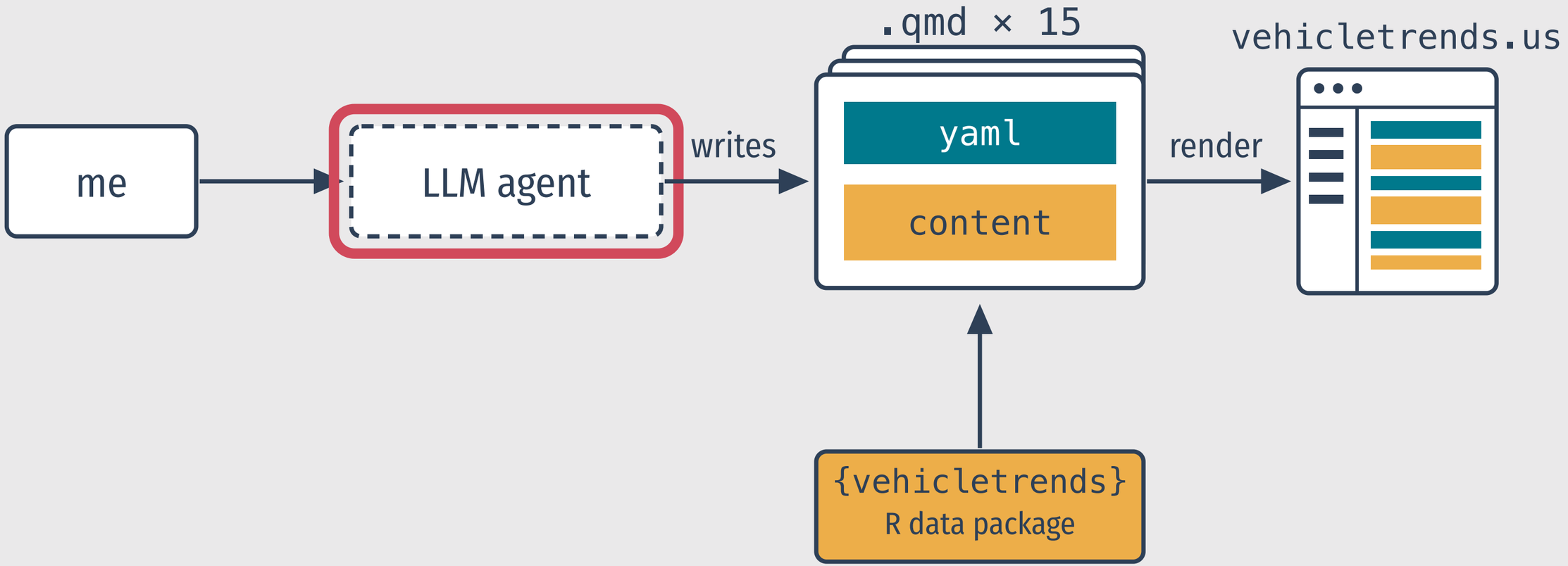
→ URLs to each page
→ URLs to each header
→ Navigation
→ Consistent layout
→ Easy deployment

_quarto.yml

```yaml
project:
  type: website

website:
  sidebar:
    contents:
      - about.qmd
      - vmt-daily.qmd
      - vmt-age.qmd
      - depreciation.qmd
      - ...
```

Charts built with {vehicletrends}


me
LLM agent
writes
.qmd × 15
yaml
content
render
vehicletrends.us
{vehicletrends}
R data package

Modularity

Efficiency

Correctness

Code chunks are all you need

.qmd × 15



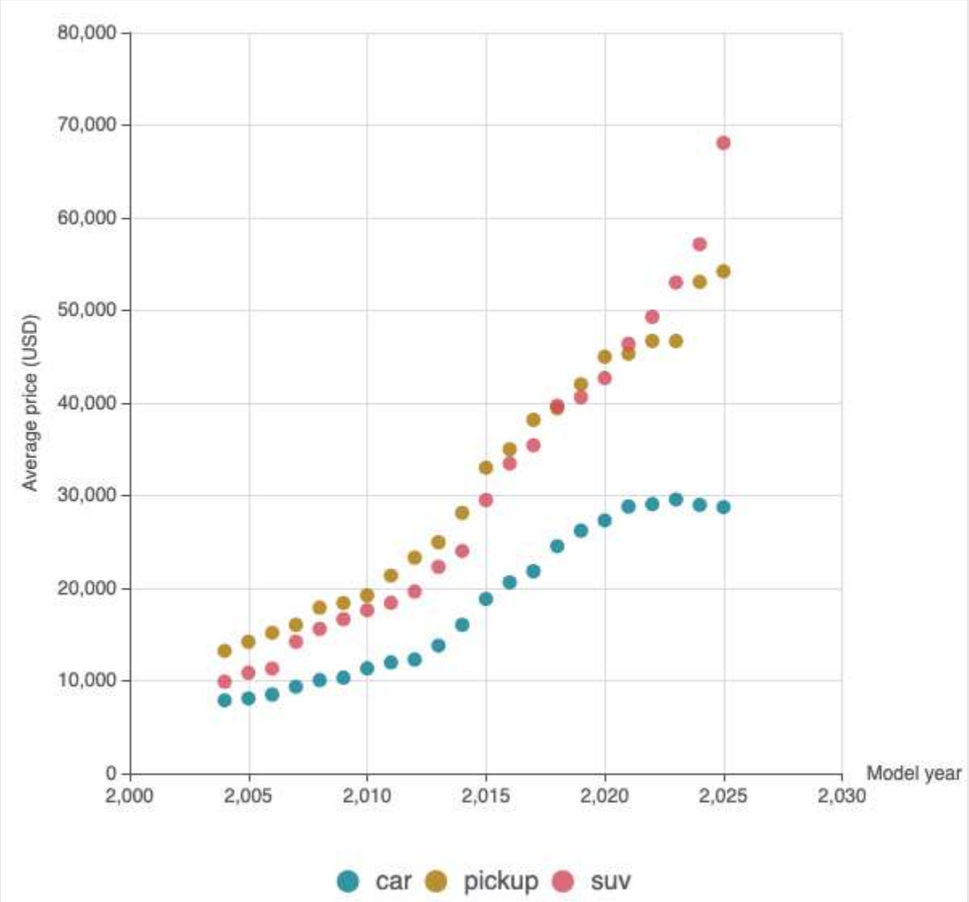
JavaScript graphics via R or Python packages

R via {echarts4r}

```r
vehicles |>
  group_by(vehicle_type) |>
  e_charts(year) |>
  e_scatter(avg_price)
```

Python via {plotly}

```python
px.scatter(
  vehicles,
  x = "year",
  y = "avg_price",
  color = "vehicle_type"
)
```



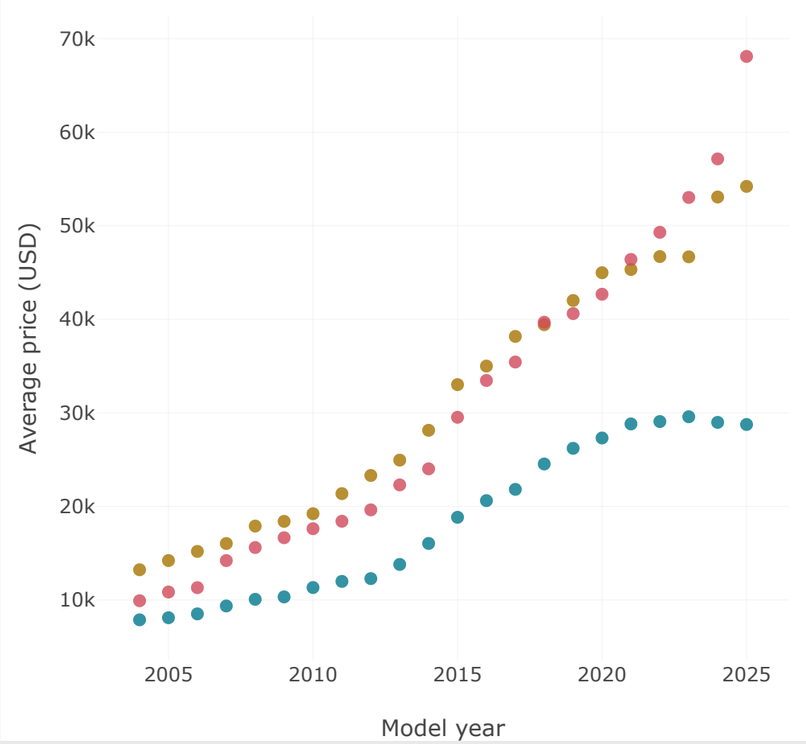
Widgets that talk to each other: `crosstalk` + `plotly`, or `ojs`

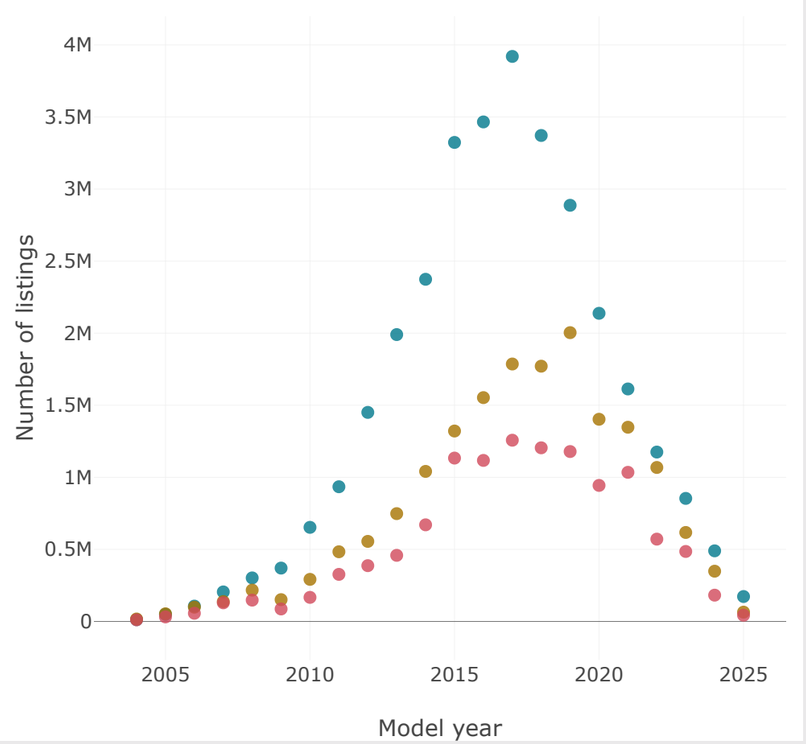
Model year

2,004 2,025

2,004 2,007 2,010 2,013 2,016 2,019 2,022 2,025

Vehicle type





Put  in a box!

Shinylive – no server

```
```{shinylive-r}
#| standalone: true
#| viewerHeight: 800

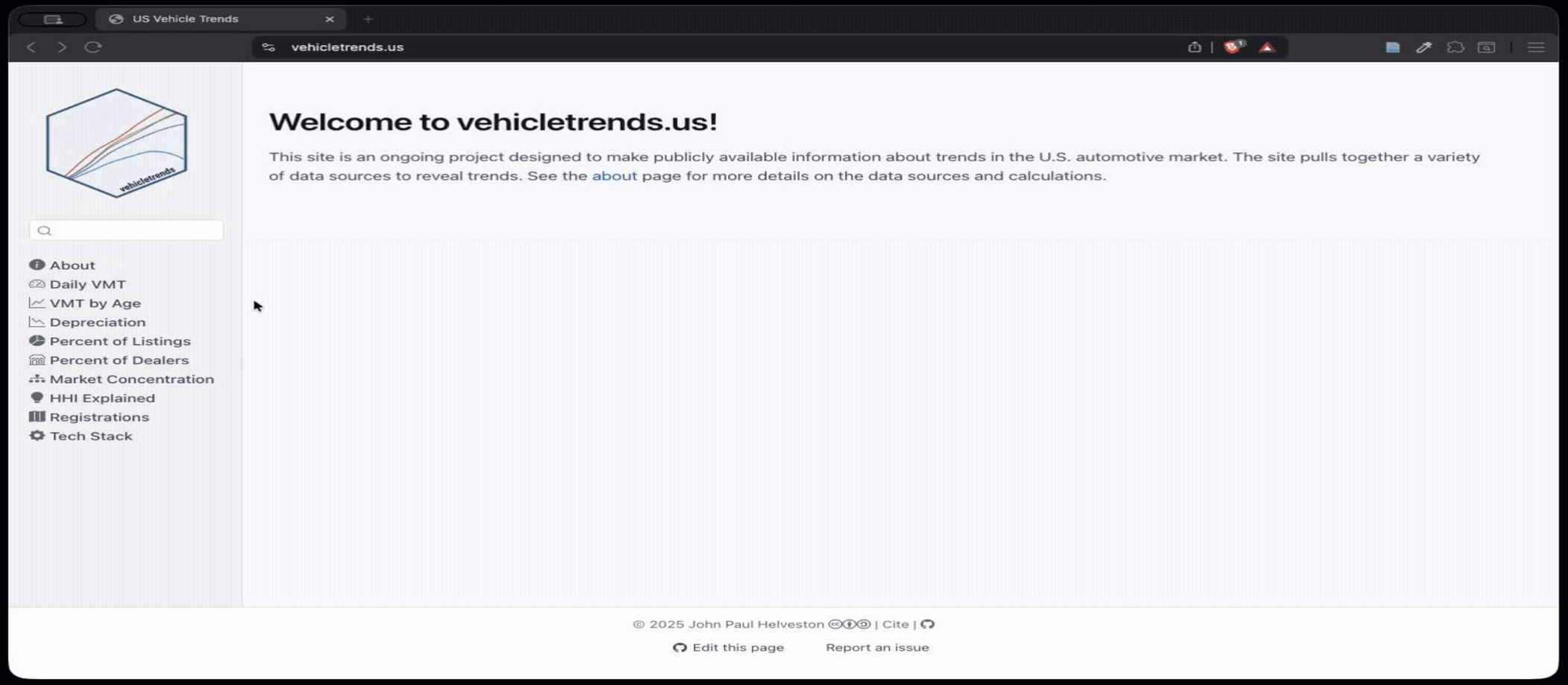
# <Shiny app code>
```
```

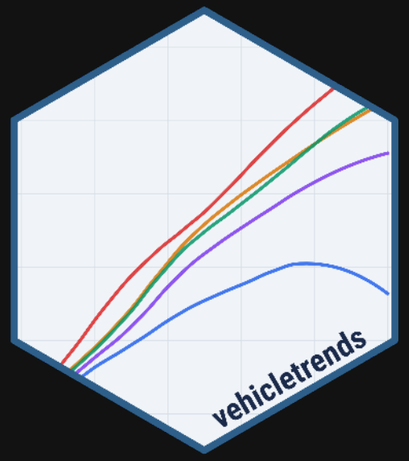
A real Shiny server

```
```{r}
htmltools::tags$iframe(
  src    = "https://app.share.connect.posit.cloud/",
  height = "700px",
  width  = "100%",
  style  = "border: none;"
)
```
```

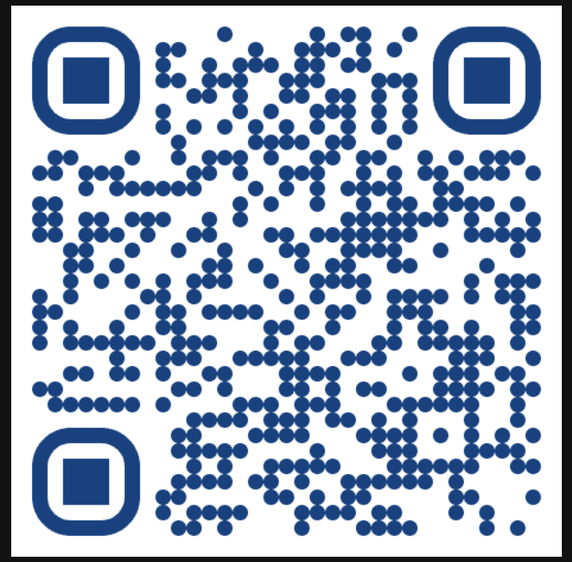
Put any widget in a box!

- **276 lines** of hand-written MapLibre
- **1.64 GB** of tiles built with `{pmtiles}` (thanks Kyle Walker!)
- Lives in it's own repo
- Hosted on GitHub pages


US Vehicle Trends
vehicletrends.us
vehicletrends
Welcome to vehicletrends.us!
This site is an ongoing project designed to make publicly available information about trends in the U.S. automotive market. The site pulls together a variety of data sources to reveal trends. See the about page for more details on the data sources and calculations.
About
Daily VMT
VMT by Age
Depreciation
Percent of Listings
Percent of Dealers
Market Concentration
HHI Explained
Registrations
Tech Stack
© 2025 John Paul Helveston | Cite |
Edit this page
Report an issue



vehicletrends





penguins dashboard



Modularity

Efficiency

Correctness

EV Sales

Sales of **electric vehicles** grew 20% in 2025, reaching over *20 million* sales globally.

EV Sales

Sales of **electric vehicles** grew 20% in 2025, reaching over *20 million* sales globally.

Quarto (.qmd)

```
---
format: pdf
---

## EV Sales

Sales of **electric vehicles** grew
20% in 2025, reaching over
_20 million_ sales globally.
```

125 characters



PDF (via LaTeX)

```
\documentclass{article}
\usepackage[T1]{fontenc}
\usepackage{geometry}
\geometry{margin=1in}

\begin{document}

\section*{EV Sales}

Sales of \textbf{electric vehicles} grew
20\% in 2025, reaching over
\emph{20 million} sales globally.

\end{document}
```

252 characters



HTML

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="utf-8">
<title>EV Sales</title>
<style>
body { font-family: Georgia, serif;
       max-width: 40em; margin: 3em auto; }
h2 { font-size: 1.6em; }
</style>
</head>
<body>
<h2>EV Sales</h2>
<p>Sales of <strong>electric vehicles</strong> grew
20% in 2025, reaching over <em>20 million</em>
sales globally.</p>
</body>
</html>
```

377 characters

EV Sales

Sales of **electric vehicles** grew 20% in 2025, reaching over *20 million* sales globally.

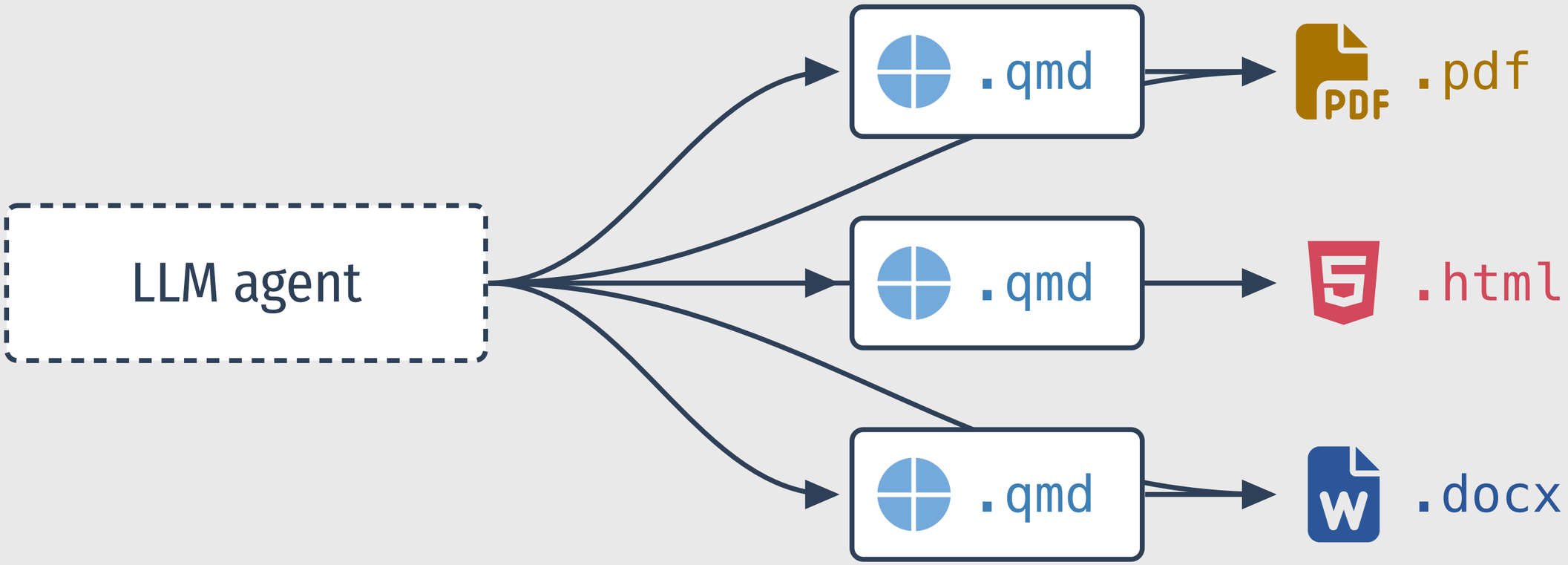


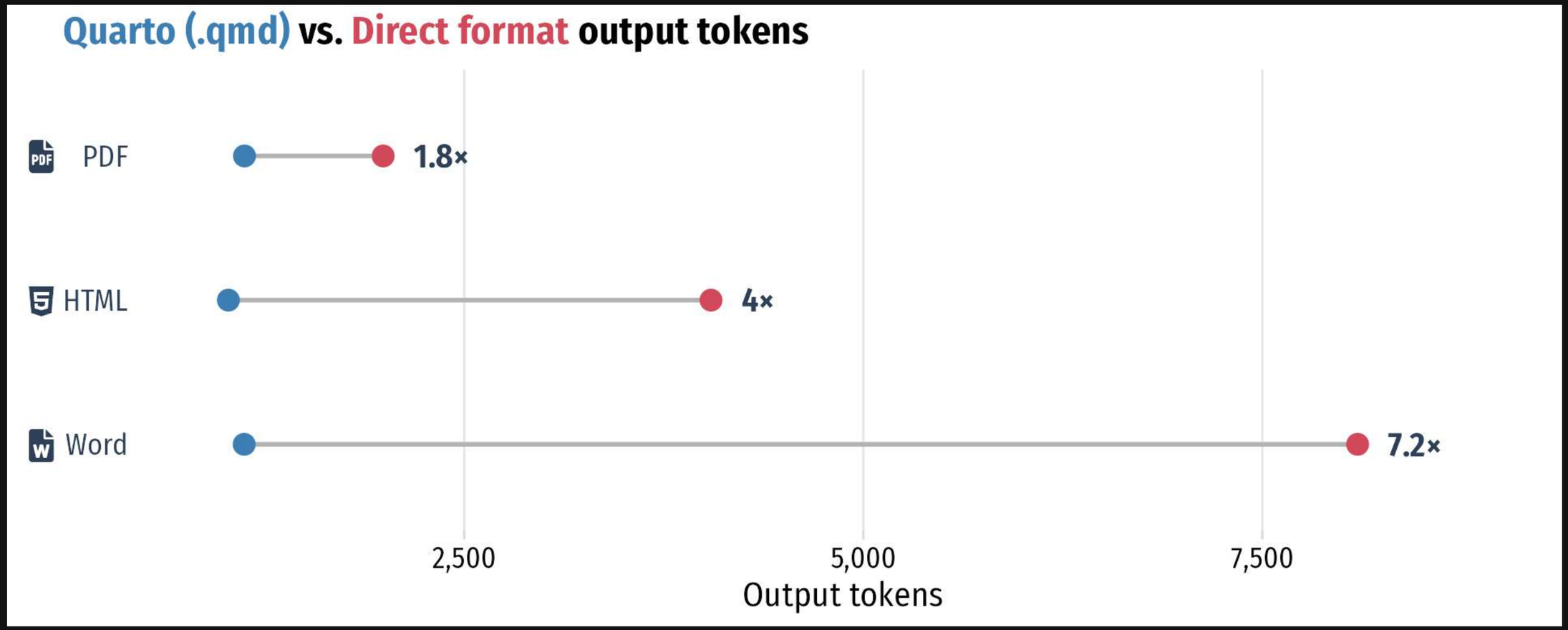
Word (`word/document.xml`)

```xml
<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<w:document xmlns:w="http://schemas.openxmlformats.org/
                     wordprocessingml/2006/main">
  <w:body>
    <w:p>
      <w:pPr><w:pStyle w:val="Heading2"/></w:pPr>
      <w:r><w:t>EV Sales</w:t></w:r>
    </w:p>
    <w:p>
      <w:r><w:t xml:space="preserve">Sales of </w:t></w:r>
      <w:r><w:rPr><w:b/></w:rPr><w:t>electric vehicles</w:t></w:r>
      <w:r><w:t xml:space="preserve"> grew 20% in 2025, reaching over </w:t></w:r>
      <w:r><w:rPr><w:i/></w:rPr><w:t>20 million</w:t></w:r>
      <w:r><w:t xml:space="preserve"> sales globally.</w:t></w:r>
    </w:p>
  </w:body>
</w:document>
```

663 characters 😱

Ask for the artifact, not the file




Quarto (.qmd) vs. Direct format output tokens
PDF
1.8×
HTML
4×
Word
7.2×
2,500
5,000
7,500
Output tokens


Always has been
Wait, it's all
just markdown?

Small pieces have *small context windows*

```
vehicletrends.us/                  lines
├── index.qmd                        7
├── about.qmd                      101
├── vmt-daily.qmd                  216
├── vmt-age.qmd                    351
├── depreciation.qmd               324
├── percent-listings.qmd           620
├── percent-dealers.qmd            199
├── market-concentration.qmd       292
├── hhi-explained.qmd              331
├── registrations.qmd              349
├── nearest-vehicles.qmd            34
├── tech-stack.qmd                  60
├── cite.qmd                        50
├── LICENSE.qmd                     26
└── 404.qmd                         21
```

depreciation.qmd

```
  ## Depreciation by body style

  Used trucks hold their value better
- than any other body style.
+ than any other body style, and the
+ gap has widened every year since 2022.
```

Small pieces have *less to agree upon*



The **input** that defines it

The **output** that consumes it

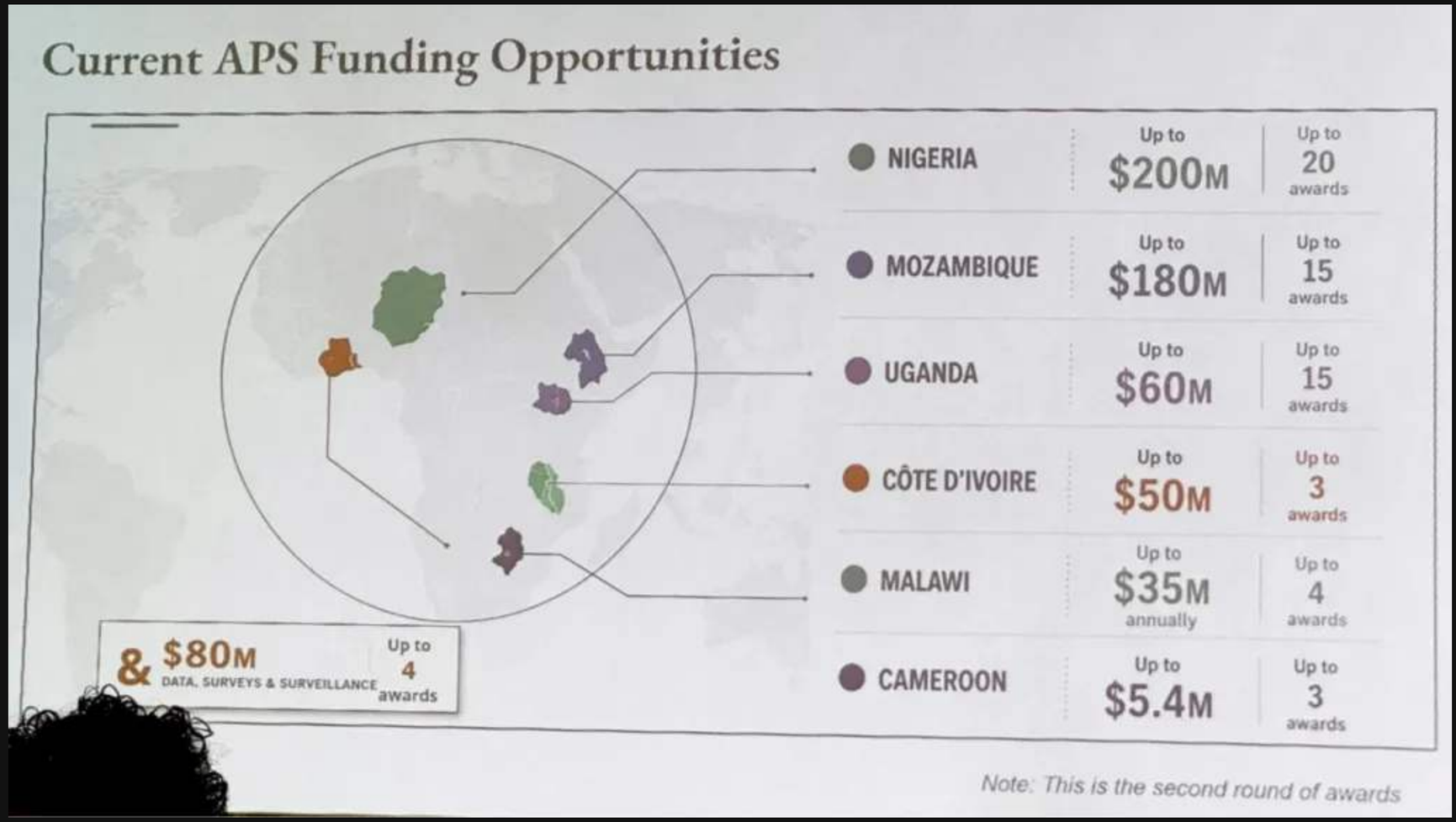
The **reactive** that connects them

The **code chunk**

Modularity

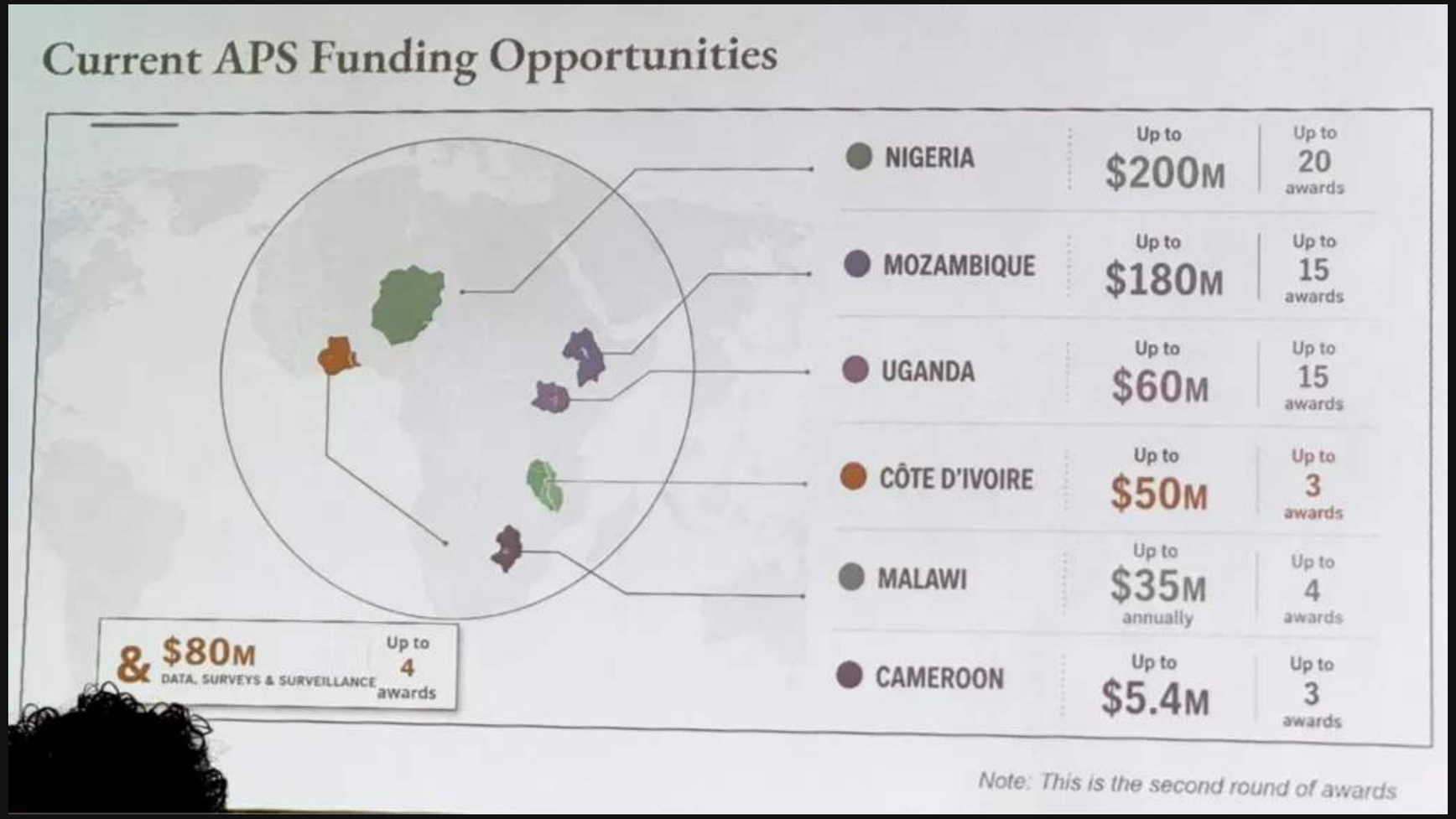
Efficiency

Correctness


Current APS Funding Opportunities
NIGERIA
Up to $200M
Up to 20 awards
MOZAMBIQUE
Up to $180M
Up to 15 awards
UGANDA
Up to $60M
Up to 15 awards
CÔTE D'IVOIRE
Up to $50M
Up to 3 awards
MALAWI
Up to $35M annually
Up to 4 awards
CAMEROON
Up to $5.4M
Up to 3 awards
& $80M DATA, SURVEYS & SURVEILLANCE
Up to 4 awards
Note: This is the second round of awards

Source: BBC

There is no version that throws an error



Source: BBC

Same map from source

```r
library(tidyverse)
library(rnaturalearth)
library(sf)

funding <- read_csv("data/africa-funding.csv")

africa <- ne_countries(continent = "Africa", returnclass = "sf")
stopifnot(all(funding$name %in% africa$name))

funded <- africa |>
  inner_join(funding, by = "name") |>
  mutate(
    lon = st_coordinates(st_point_on_surface(geometry))[, "X"],
    lat = st_coordinates(st_point_on_surface(geometry))[, "Y"]
  )

ggplot(africa) +
  geom_sf(fill = "grey90", color = "white", linewidth = 0.2) +
  geom_sf(data = funded, fill = "#00798c", color = "white") +
  geom_segment(
    data = funded, color = "#2e4057",
    aes(lon, lat, xend = lon + dx, yend = lat + dy)
  ) +
  geom_text(
    data = funded, color = "#2e4057", size = 5, lineheight = 0.9,
    family = "Fira Sans Condensed",
    aes(
      lon + dx, lat + dy, hjust = ifelse(dx > 0, -0.05, 1.05),
      label = paste0(name, "\n$", amount, "M")
    )
  ) +
  coord_sf(xlim = c(-38, 66), ylim = c(-36, 38)) +
  theme_void()
```



Architect robustness into the workflow



Modularity
Efficiency
Correctness

Not just dashboards

Reports & manuscripts

Citations, cross-refs — the apparatus of academic writing

One template, every participant

Parameterized reports, fanned out per cohort or condition

This talk

A `lexis-revealjs .qmd`, rendered

Same architecture, whatever the audience needs to trust the numbers.

“The Unreasonable Effectiveness of Quarto”

Quarto wasn’t built for AI!

vehicletrends



Thanks!

jhelvy.com

jph@gwu.edu

@jhelvy

penguins dashboard

